METAL DETECTOR

A Dissertation
Presented to
The Academic Faculty

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ACRONYMS

SDR …….Software Define Radio

USRP…… Universal Software Radio Periphery

VNA ……..……..... Vector Network Analyzer

NI ……...National Instruments

GPR …….Ground Penetrating Radar

TX ……Transmitter

RX …….Receiver

GHz ……...Giga Hertz

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SUMMARY

The world has become more vulnerable to terrorism in the last few decades. The common people have become soft target for terrorist organizations these days. The terrorist of these organization attack public places and large gatherings so that large number of people can be victimized. Moreover, number of suicide bombing at public places has risen significantly in past decade. Therefore, the security of the people in presence of such terrorist activities is quite difficult for the law enforcement agencies especially in the developing countries because they usually lack the latest equipment to prevent mass slaughter in form of suicide bombing and other terrorist activities. A long range metal detector and a Ground Penetrating Radar (GPR) is made in this project that can detect the suicide bomber or an armed terrorist from a distance of 10 to 15 meters and the GPR would be able to detect underground mines up to three feet by varying frequencies. For this aspect horn antennas of high gain and directivity are designed using HFSS software to transmit and receive RF waves using USRP-2901. The reflected waves are analyzed using LabVIEW's toolkits and the strength of the reflected signal used to differentiate the perspective threats.

CHAPTER 1. INTRODUCTION

Security of its citizens is always being a major issue for every country; especially in the developing countries providing security to its citizens is quite challenging task for government and law enforcement agencies. These security issues are mainly due to the scarce resources which law enforcements agencies face in developing countries. The scanning devices these agencies are using are quite obsolete and do not perform their tasks effectively. Due to this the developing world is more vulnerable to terrorism then the developed world. Since, the inception of terrorism the developing countries are facing more causalities then the developed world. The fundamental reason for this problem is that effective security scanning devices are not locally made in developing countries like Pakistan. These devices need to be imported from developed word and are hence quite expensive. Moreover, these scanning devices are not being designed according to the required need and vicinity of the developing world countries.

Suicide bomber are also one of the challenging part of this terrorism. These bombers wear explosive vests that contain explosives and metals (metal ball bearings) to achieve the target of mass destruction and massive causalities. These suicide bombing initiated in 1981 and from that till 2015 this type of terrorism have killed 50000 peoples across 40 countries. Similarly, the global suicide attack rate increase to one a day from 2003 to 2015 from one a week in 2001 to 2003 and one a month in 1980's to 1990's. [1] Similarly underground mines is a huge problem for the law enforcement agencies specifically in the areas where they are fighting guerilla war against these terrorist.

Therefore, for solving this problem for the developing world and more specifically for Pakistan we are designing a long range metal detector and a GPR (Ground penetrating radar). This metal detector/GPR can be used by law enforcement agencies to eventually increasing the efficiency of security at highly sensitive areas. The metal detector would be able to detect any perspective threat with at least fifteen to twenty meters' distance and GPR could be used to detect any metal or mine underground using different frequencies. Hence, it can also be used to detect suicide bombers and underground mines. This device would be designed according to requirement of our country and made it quite cost effective so that other developing could also get benefit out of it too.

1.1.1 Initial Study

In the initial study we had found the existing devices used for metal detection and potential suicide bomber. There are different kind of devices some detect the potential threat with a very less distance and some detect with a quite moderate distance of 100 meters. CBD-1000 Concealed bomb detection system is a device designed for detection of objects that are being used for suicide bombing. It can detect the threat up to a distance of three meters. The other quite effective device is counter bomber that detects the threat to good distance of hundred meters. [2]

Metal detection can be done in many ways one of the conventional method for metal detection is X-rays. X-rays are considered to be quite injurious to health. But using X-rays from a distance of ten to twenty meters decrease its adverse effects on human. Thus X-rays can be used for metal detection a transmitter transmits the X-rays that

requires a detector which is mostly a plastic sheet that is monitored by a camera which had a wireless link to data analysis base. [3]

Cloths and other materials seems approximately transparent as the radiations wavelength increase to terahertz range. They have wavelength larger than 300 micrometers and have a frequency of about 1 THz. Imaging in this range can detect explosives carrying humans under their cloths without the ionization of radiations. One of the practical and low cost sources for producing these waves is quantum cascade lasers. These small semiconductor lasers have the ability to operate at as low as 1.5 THz frequency. Microwaves can also be used for imaging and the detection metals in the frequency band of 100GHz to 300GHz. [4]

The conventional way for detecting metallic objects is done by and held metal detectors that are working on the magnetization phenomena. In this method magnetic field is produced between two conducting coils by passing a DC voltage at high frequency and detect the variation in magnetic field caused due to metallic objects passing between them. This method can be commonly seen in walk through metal detecting doors. Another application of this concept can be observed in hand held metal detectors (used by security guards) in these detectors there is a single conducting coil which transmits magnetic field for short interval (half second) and then receives the reflected field (other half second) and then measures the difference in both signals. These methods are most effective to use but their range is very limited the walk through door method can extend the detection range to about one and a half meter whereas, the handheld detector has a minute range of about five centimeters.

1.1.2 Our Methodology

We are using RF waves for detecting metals from a large distance. We are using a SDR of National Instruments USRP-2901 for generating RF waves and their signal processing in LabVIEW. We are using directive antennas for focusing RF waves beam in a particular direction and for getting reflection having good magnitude that can help us in signal processing. Horn antennas are the best directive antennas that can be locally made so we designed them using HFSS software and made their physical structure.

Following are the block diagrams of our project:

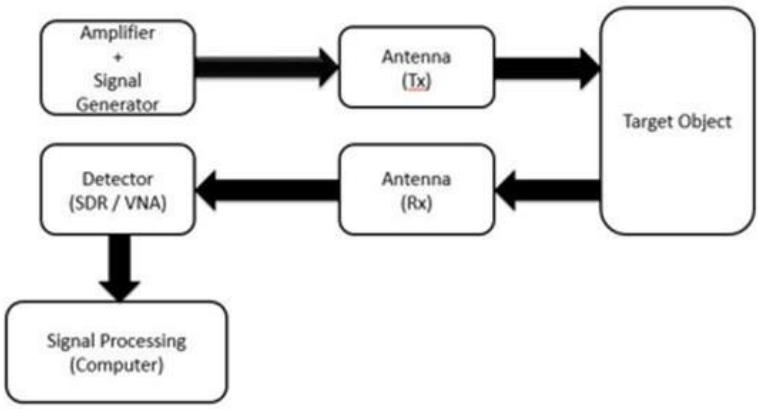


Figure 1: Complete Project Block Diagram

Block diagram for transmitter and receiver circuits are as follows:

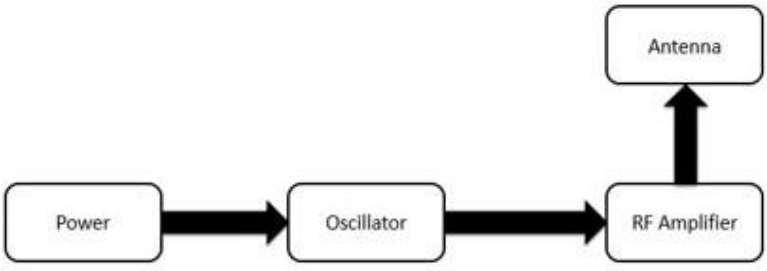


Figure 2: Transmitter Block Diagram

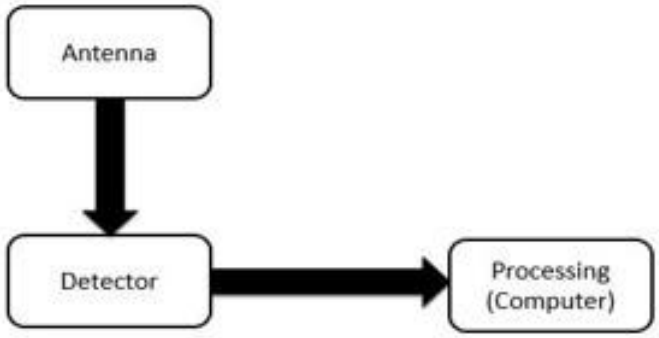


Figure 3: Receiver Block Diagram

CHAPTER 2. LITERATURE REVIEW

Based on the prior studies and the research taken from the thesis of Sara Smith's paper on "Concealed Explosives Detection using Swept Millimeter Waves" we found out that weapons/metal detection is an active research topic and by now many of the security organizations are now funding for this field of research many of these organizations are linked to the aviation security. Passive emission in the millimeter wave band can be used to form images of people which can be used for security screening. Differences in emitted brightness between the human body and a concealed object provide the contrast in the image. Basically the active imaging uses the radiations and the waves that are reflected from the body to form an image of the target body then the difference in reactivity of the body and any metal are used to form an image from which then indicates the presence of the metal. As less power levels are used in active system so the emission of the electromagnetic radiations is also negligible. [5]

Also Fabrey-Perot type technique can be used for the detection of the metals in free space. This method doesn't generate images eventually surpassing the privacy concerns associated with image-forming systems and is also less expensive. In this method the Fourier transform of the reflected wave data is used to detect the optical thickness of the target material. The Fourier transform assumes that the data is periodic and this technique is used at gigahertz frequencies as long as the sweep is sufficiently large. [6]

The basic techniques and the initial study of the Ground Penetrating Radar were taken from the "Ground Penetrating Radar as a Subsurface Environmental Sensing Tool" by LEON PETERS JR. We found out that the GPR are used as the environmental tools and the basic

phenomenon involved the antennas, propagation target scattering and mapping. It can be used to identify metallic pipes and the detection of the land mines and the other metallic elements is also one of the serious environmental concerns. The basic GPR includes a signal generating unit and the antennas are used to send the signals into the ground sometimes it also has an internal storage unit to store the data for the examination after the fieldwork. The antenna amplifies the signal and then transmits it into the ground at a certain frequency. The higher the frequency of the antenna the shallower it will penetrate into the ground. The reflected wave from the target material is also received by the antenna and then the processing is done. The amplitude of the reflected signal is observed and then commutations are made accordingly. As metals are considered to be the complete reflector so don't allow the wave to pass through and the whole wave is reflected. [7]

CHAPTER 3. WORKING

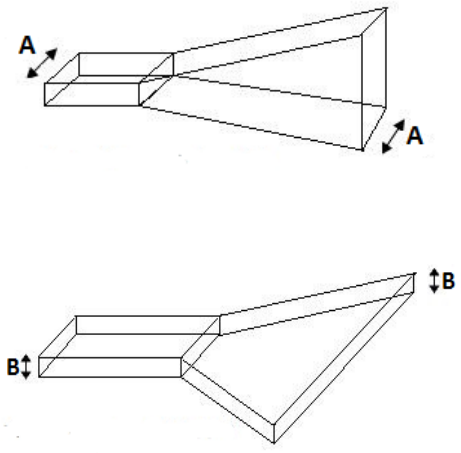
The major portion of project is research based so initially we have decided to simulate the project using the software defined radios (SDR) and antennas. The antenna of center frequency will be 2.1GHz was designed on Ansoft's HFSS(as discussed below).

A signal wave will be generated and transmitted by a software defined radio and will be transmitted through an antenna and the same wave will be received after striking the target object. Then transmitted and received signals will be analyzed by using the gnu radio software (for SDR). Basically, we will observe the difference between the received signals when they strike a metallic object and when they strike a non- metallic object and based on the difference between the two we can distinguish between them.

CHAPTER 4. ANTENNA DESIGN

4.1.1 Antenna Design

Horn antennas are widely used in application where high gains and high frequencies are involved, it is to be said that the gain of a horn antenna increases as the frequency increases and the beam width decreases with an increase in frequency. Designing of a horn antenna is challenging as it involves the detailed calculation of both the E-Plane (horizontal) represented as “A” and H-Plane (vertical) dimensions represented as “B” in the diagrams below. Also a Pyramidal horn antenna is flared on both planes at the end and the dimension of the flare with the “x” and height “y” having the also needs to be calculated precisely.



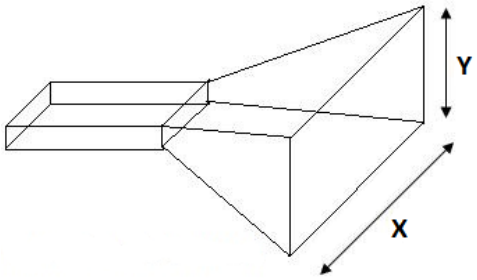


Figure 4: Horn Antenna Designs

Typically Horn Antennas are fed by using a waveguide it is basically a simple metal cavity which guides the electromagnetic energy from on point to another waveguide is fed with a small dipole usually of length λ/4. The diagram below shows the dimensions of the waveguides with width "V" and height "U" and the feed element is represented in red color.

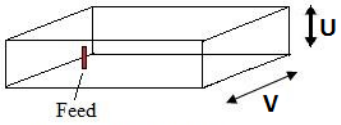


Figure 5: Horn Antenna Waveguide

A horn antenna is a waveguide structure which may be pyramidal or conical. Horn antennas are widely used as antenna as ultrahigh frequency above 300MHz and is being used to transmit radio waves out to the space. The antenna was designed using the Ansoft's HFSS simulation software.

A horn antenna is a waveguide structure which may be pyramidal or conical. Horn antennas are widely used as antenna as ultrahigh frequency above 300MHz and is being used to transmit radio waves out to the space.

We have used a standard waveguide WR430 which has frequency range of 1.7GHz to 2.6 GHz. So, we only had to calculate the horn flare dimensions. For this we used the basic equation for designing a horn antenna is the horn design equation:

$$\left(\sqrt{2x} - \frac{b}{\lambda}\right)^2 (2x - 1) = \left(\frac{Go}{2\pi}\sqrt{\frac{3}{2\pi}}\frac{1}{\sqrt{x}} - \frac{a}{\lambda}\right)^2 \left(\frac{Go^2}{6\pi^3}\frac{1}{x} - 1\right)$$

Where

G_o = Gain

λ = Wavelength

x = trial value

a = horn flare width

b = horn flare length

A horn antenna design is valid only is the horn antenna equation is satisfied. And the value of x-trial mentioned is formula is a constant value to balance the equation and can be found as:

$$x(trial) = x_1 = \frac{Go}{2\pi\sqrt{2\pi}}$$

Moreover, the length of horn is determined by calculating E- plane length p_e and H-plane length p_h. And for a valid design these should be equal. The formulas for calculating them are:

$$p_e = (b_1 - b)\left[\left(\frac{p_e}{b_1}\right)^2 - \frac{1}{4}\right]^{1/2}$$

$$p_h = (a_1 - a)\left[\left(\frac{p_h}{a_1}\right)^2 - \frac{1}{4}\right]^{1/2}$$

Where,

$$a_1 = \sqrt{3\lambda\rho_2} \cong \sqrt{3\lambda\rho_h} = \frac{G_0}{2\pi}\sqrt{\lambda\frac{3}{2\pi X}}$$

$$b_1 = \sqrt{2\lambda\rho_1} \cong \sqrt{2\lambda\rho_e} = \sqrt{2X\lambda}$$

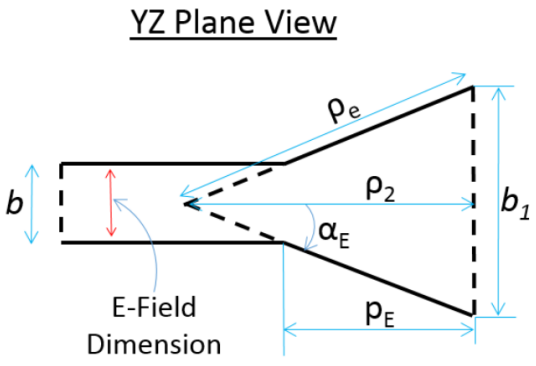


Figure 6: Antenna Diagram

$$\frac{\rho_e}{\lambda} = X$$

$$\frac{\rho_h}{\lambda} = \frac{G_0^2}{8\pi^3}\left(\frac{1}{X}\right)$$

Formulas taken from Blanis's book.[8]

4.1.2 Calculations of Parameters Using MATLAB

The calculations of the antenna parameters were done of MATLAB and a code was designed to get the output parameters:

```
clc
v=3e8;
f=2.1e9;
a=4.3;
b=2.15;
l= (v/f)*39.37

ga=1.5;
g=10^ga

x= g/((2*pi)*(sqrt(2*pi)))

Pe = x*l
Ph = ((g^2)/(8*(pi^3)))*(1/x)*(l)

right = ( sqrt(2*x) - (b/a) )^2 * ((2*x) - 1)
left = ( (g/(2*pi) * sqrt(3/(2*pi)) * 1/sqrt(x)) - a/l )^2  *  ( ((g^2/(6*(pi^3))) * 1/x ) - 1 )

A= sqrt(3*l*Ph)
B= sqrt(2*l*Pe)

Pei = (B-b) * sqrt(((Pe/B)^2) - 1/4)

Phi = (A-a) * sqrt(((Ph/A)^2) - 1/4)
```

The above given MATLAB code can be used to design a horn antenna.

4.1.3 Output of MATLAB Code

```
                    l =

A =                     5.6243

   13.8036
                    g =

                       31.6228
B =

   11.2706          x =

                        2.0078
Pei =
                    Pe =
    7.9193
                       11.2927

Phi =               Ph =

    6.1538             11.2927
```

4.1.4 Probe Feeding

This part basically tells about the probe feeding part of the pyramidal horn antenna, horn antenna is fed from the waveguide; transmission is done through coaxial cable and waveguide in TEM mode and TE mode respectively, this is done to get low VSWR minimum return loss and prefect impendence matching, the extended part of the pin inside the waveguide acts as a monopole.

Here “L” is the length of the inner extended part and “D” is the distance of the extended part from the closed end of the waveguide.

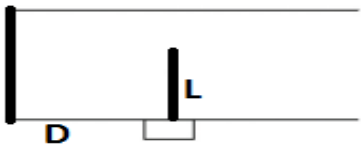


Figure 7: Probe Dimensions

Many simulations were performed on Ansoft's HFSS to obtain the desired results, to find the exact position of the feed, calculations for $\boldsymbol{\lambda_g}$ (cut-off frequency wavelength) and $\boldsymbol{\lambda_o}$ (free space wavelength) were performed.

$$\boldsymbol{\lambda o = \frac{c}{f} = \frac{3.0 \times 10^8 m/s}{2.1 \times 10^9 Ghz} = 14.2 cm}$$

$$\boldsymbol{\lambda g = \frac{1}{\sqrt{\frac{1}{\lambda_0^2} - \frac{1}{\lambda_c^2}}}}$$

Now the transmission in higher mode is also dependent on the dimensions of the waveguide and the wavelength can be calculated by the following formula.

$$\boldsymbol{(\lambda_c)_{mn} = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}}$$

The first mode of transmission is TE10 (m =1, n = 0), thus giving (λc) 10= 2α = 21.844 cm. By introducing λ_o and λc into (1),we obtain λg = 11.90 cm. Therefore the depth of waveguide λg /2 is equal to 5.95 cm

So the position and the distance of the feed from the edge of the waveguide will be now λg /4 = 2.97cm inside the waveguide and the height of the feed will be equal to λo /4 = 3.55cm.

4.1.5 Designing in HFSS

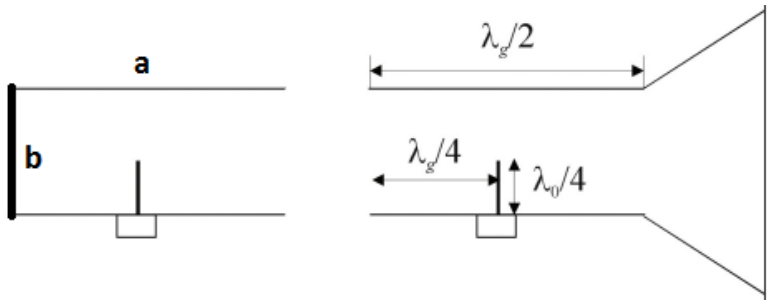


Figure 8: Antenna Dimensions

Now for the horn antenna's E-plane aperture and H-plane aperture are found using:

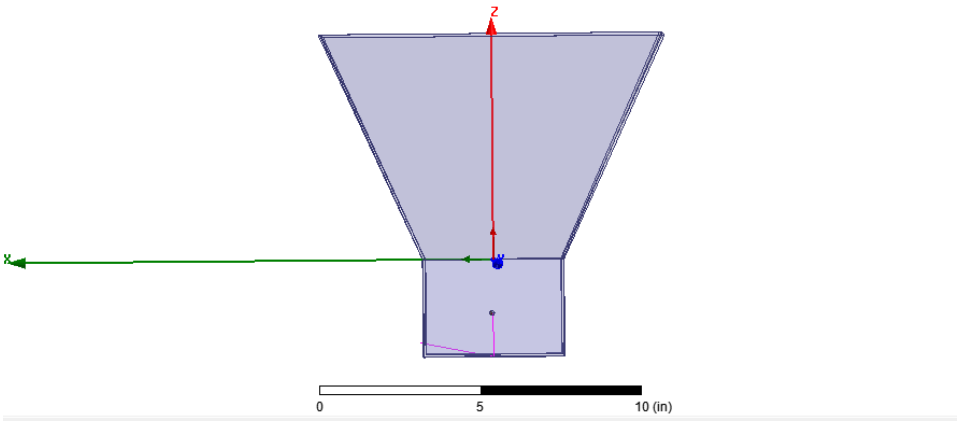


Figure 9: Horn Antenna in HFSS

A gradual mathematical as well as practical approach was followed for the designing of the horn antenna.

The dimension of the Antenna was as follows:

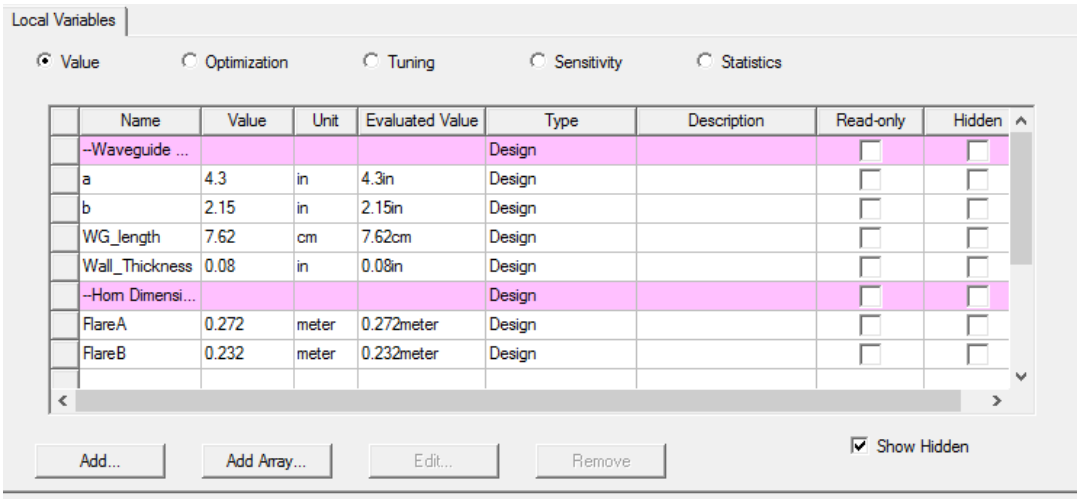


Figure 10: Dimensions of Horn Antenna in HFSS

The output plot of s-11 parameter of the antenna was also observed using the software and shown in figure below:

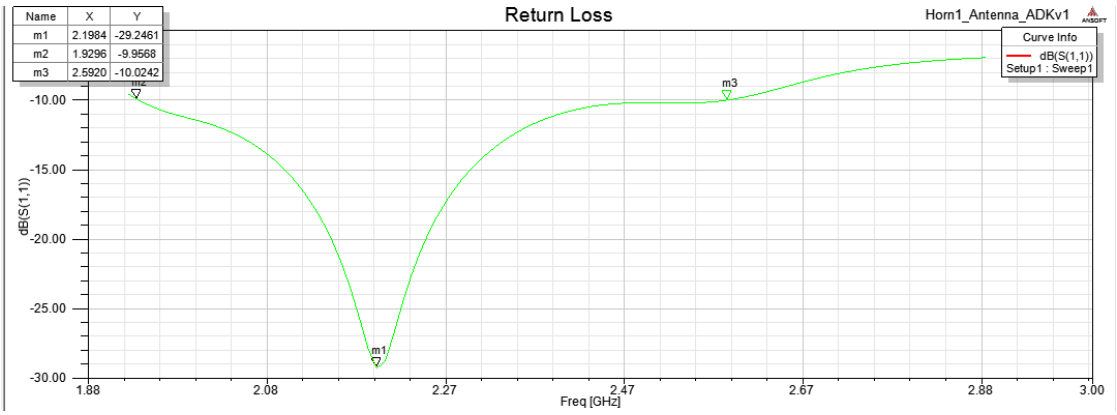


Figure 11: S11 Output of Antenna

CHAPTER 5. METAL DETECTION IN FREE-SPACE

This part is one of the main and important part of our final year project. Metal detection in free space is actually our proof of concept that we can detect metal from long range using our setup. Its efficiency is the main issue which can be easily improved to a very accurate level using highly calibrated SDR's. In our case we don't have calibrated SDR so our efficiency of long range metal detection had effected.

5.1 The Setup

Our setup consist of two our own designed and fabricated horn antennas. Two SDR's of National Instruments namely USRP-2901. Two Laptops with LabVIEW 2014 and images of USRP-2901 installed. One pair of laptop, horn antenna and USRP is acting as transmitter and the other pair is acting as receiver.

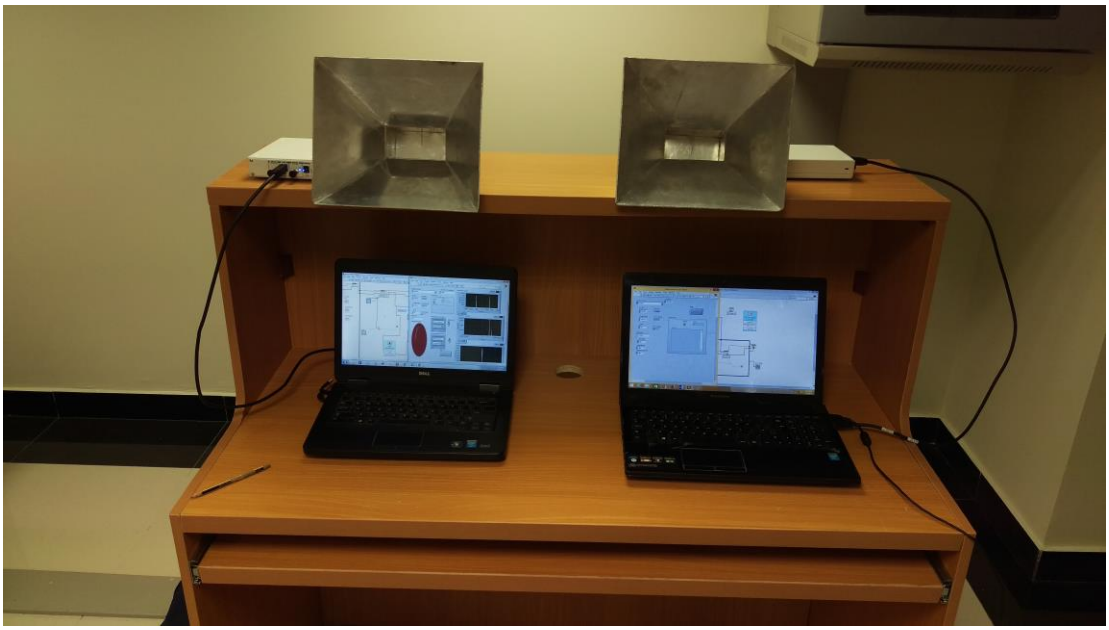


Figure 12: Our Setup

5.2 Working

In the process of metal detection two NI USRP 2901 were used as a signal generator and the NI LabVIEW was used as the platform for the designing of the interface. One USRP is used for transmitting the signal and the other is used to receive the same signal after it has struck the target object such a configuration is known as half-duplex method. Two separate USRPs are used to eliminate the chances of leakage of transmitted signal to the receiver circuit because the receiver circuit can only bear a signal of -5 dBm whereas the signal transmitted is powered to maximum gain (20 dBm). Therefore, two USRPs are used for isolation purpose and prevent our system for permanent damage of USRP circuitry.

5.2.1 Transmitter:

The transmitter interface was rather very simple. The transmitter interface was designed in the LabVIEW and both the block diagram and the front panel is described below.

Transmitter Block Diagram:

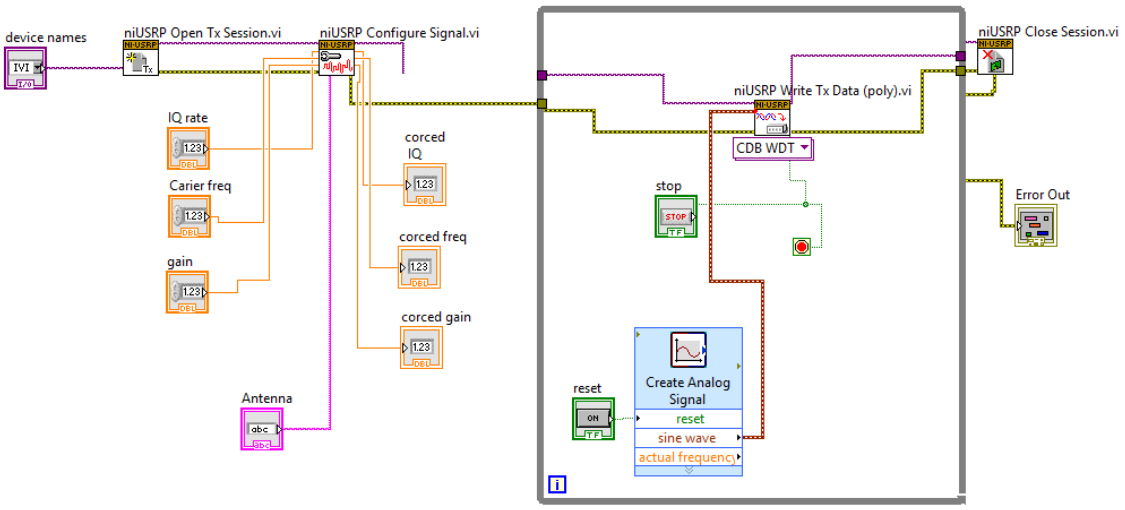


Figure 13: Transmitter VI in LabVIEW

Initially an Open Tx Session block is imported for the initialization of the code then in the niUSRP Configure Signal block we set the input blocks of IQ rate, carrier frequency and gain. Similarly blocks for coerced frequency, coerced IQ rate and coerced gain are attached that will tell us the actual values of frequency, gain and IQ rate on our GUI. Then we used the while loop block that have a niUSRP Write Tx Data block which actually write data on the USRP which finally transmit it. A signal generator block is used in the loop for sine wave generation. This wave is then given to Write Tx Data block. After the loop a close session block is used which ends the session.

Here is the transmitter's Front Panel in LabVIEW

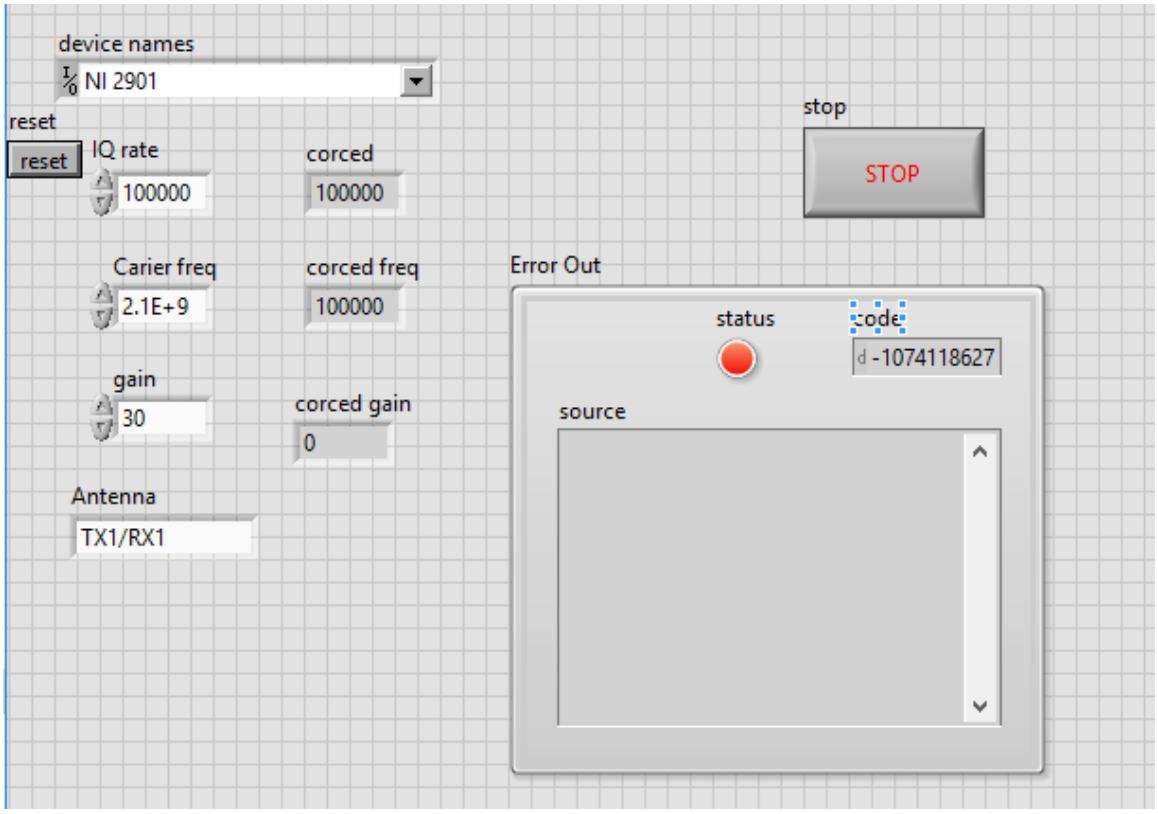


Figure 14: Transmitter Front End in LabVIEW

5.2.2 Receiver

The reflected signal from the metal is then processed using the LabVIEW interface and the peak detection algorithm is used to process the received signal.

Receiver Block Diagram:

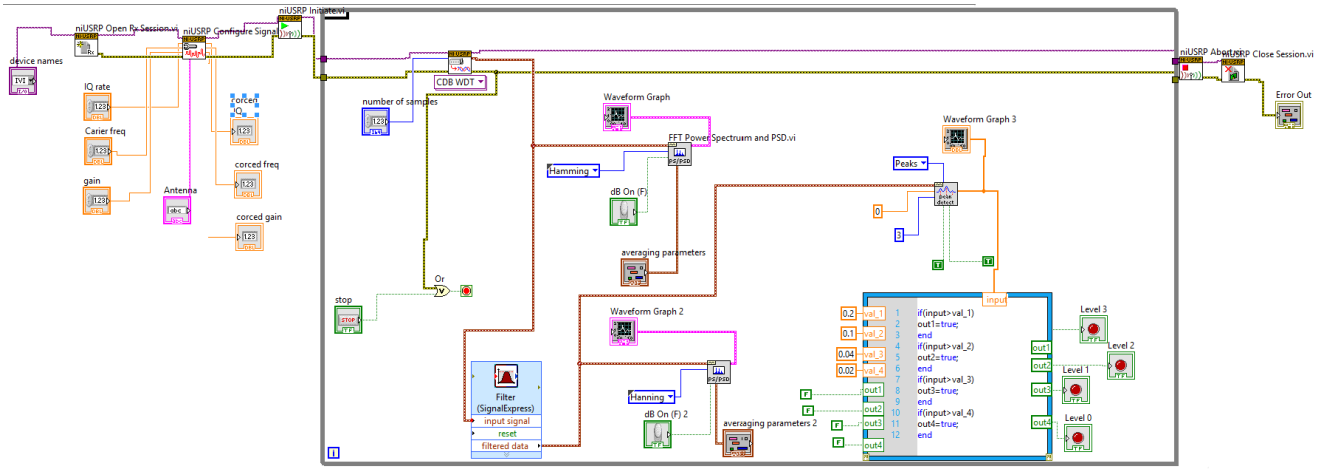


Figure 15: Receiver VI in LabVIEW

In the receiver we used the niUSRP Open Rx session. Then we have a block of niUSRP configure signal that have blocks for setting the IQ rate, carrier frequency and gain of the receiving signal. Then we have niUSRP read Rx block which takes the data of the received signal from configure signal session. Then the received signal is given to a filter block. After filtering the signal is given to the peak detector block which give us magnitude of the received signal. Then this value of magnitude is given to the matlab block where we had write an algorithm. This algorithm compare the values of magnitudes from peak detector with pre-defined values for different threats. After comparing algorithm on the LED's when a specific received signal's magnitude matches with the pre-defined threat's magnitude.

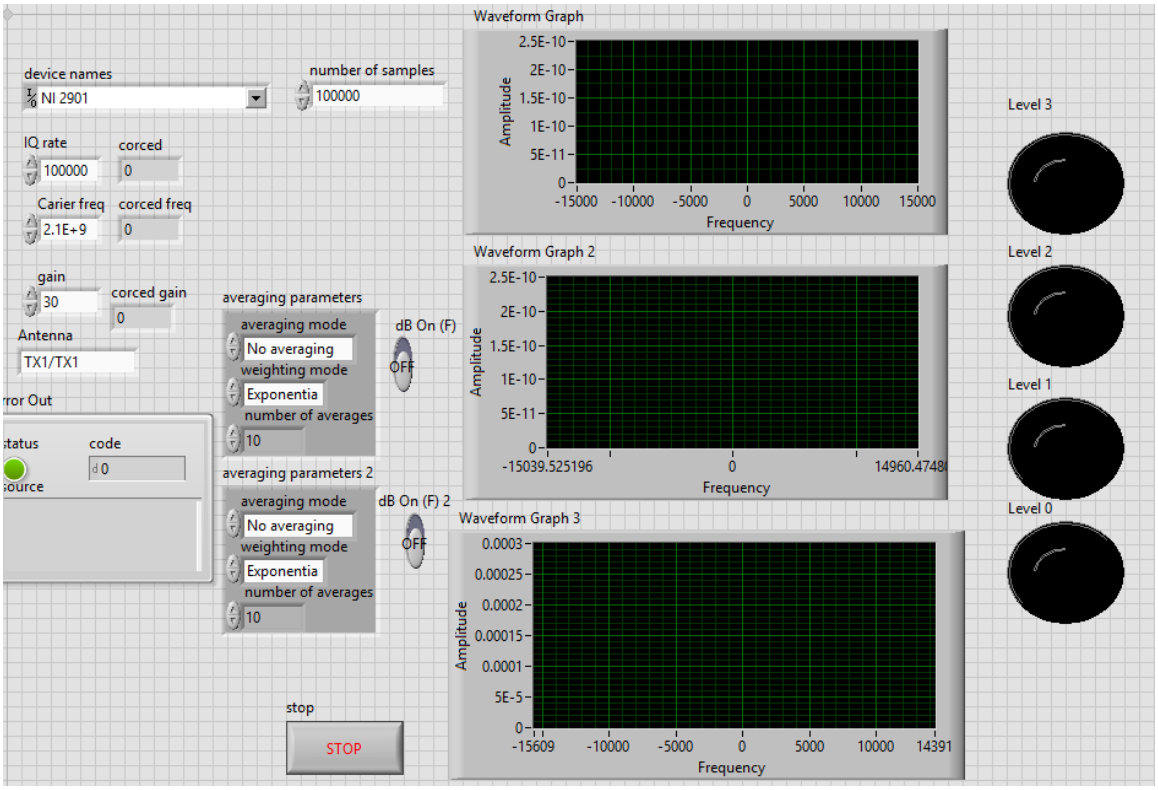


Figure 16: Receiver front end in LabVIEW

CHAPTER 6. GROUND PENETRATING RADAR (GPR)

Our current scheme of detecting metal in free space can also be used as a simple Ground Penetrating Radar (GPR) by applying the above detection scheme in full-duplex instead of half –duplex. GPR is a system used to detect underground objects/minerals in an area and can also be used to detect underground pipelines and can even detect leakages in pipelines. Through GPR system we can successfully detect and locate mines thus saving lives of people in the area.

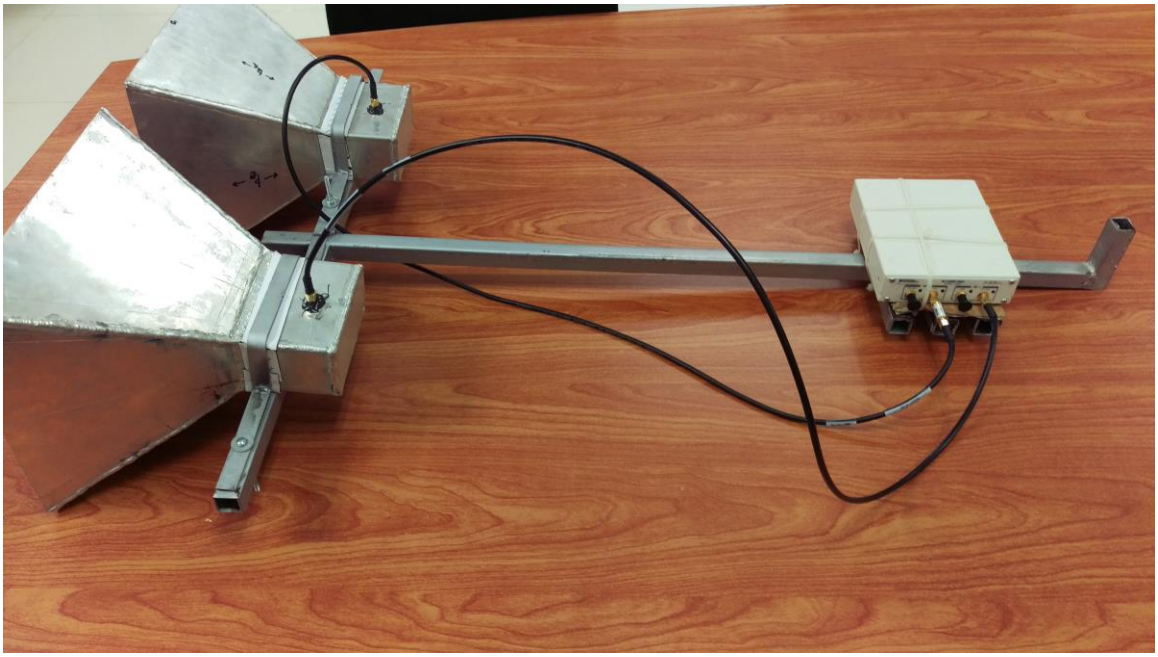


Figure 17: GPR Physical Structure

6.1.1 Basic Scheme

The scheme we have applied to make a GPR is same as we have discussed in detection of metallic objects in free-space but in here we are transmitting the signal in the ground and

measure the amplitude of reflected signal and make our judgment based on the reflected amplitudes. The amplitude of reflected wave depends upon the dielectric constant and conductivity of the material through which the wave is passing.

6.1.2 Antenna Selection

Lower frequencies provide more penetrating ability but have less resolution whereas; higher frequencies offer more resolution but less penetrating ability. Therefore, selection of antennas depends upon the desired depth and signal resolution. If materials to be analyzed are buried deep below ground then antennas operating at lower frequencies are advised but it would lack resolution ability similarly, if materials/objects to be detected lie close to ground level then antennas operating at higher frequencies are advised.

Since, we already have designed the antennas of 2.1GHz so we decided to use them. 2.1GHz is relatively a high frequency therefore; the penetrating ability of the transmitted signal will be relatively low around 1 to 3 feet.

6.1.3 Coding

In order to reduce the weight of system for better mobility we are using only one USRP-2901 to transmit and receive signal. This phenomenon of transmitting and receiving same signal using same device simultaneously is known as full-duplexing. Therefore, for GPR we are using full-duplex scheme.

For programming a full-duplex scheme in LabVIEW we simply edited the full duplex example code in software according to our requirements. In this code both transmission and receiving sessions are run in parallel and we utilize both channels of USRP-2901 (i.e.

channel 0 and channel 1) one channel is used for transmitting the signal and the other is used for receiving the signal.

For transmitting a signal we have to first initiate the by using “niUSRP Open TX session .vi” block and add the session parameters like USRP Enabled Channel (Channel 1 for transmission), IQ rate(100000), gain and active antenna (TX/RX1). The block diagram for initializing transmission is shown below:

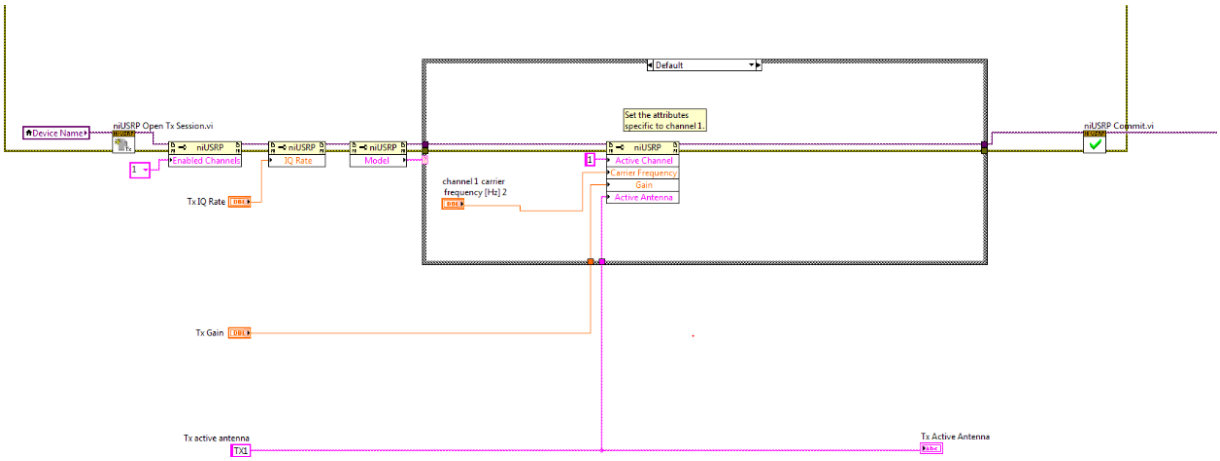


Figure 18: Initializing Tx session

After initializing the signal we have to transmit our data through antenna and then the transmission session is ended by “niUSRP Close Session.vi”. The data is transmitted using a continuous while loop which can be interrupted using a stop button on front panel. In the loop we are generating a simple sine wave of 10 kHz frequency and are transmitting it through “niUSRP Write Tx Data (poly) .vi” block. The code for this session is shown below:

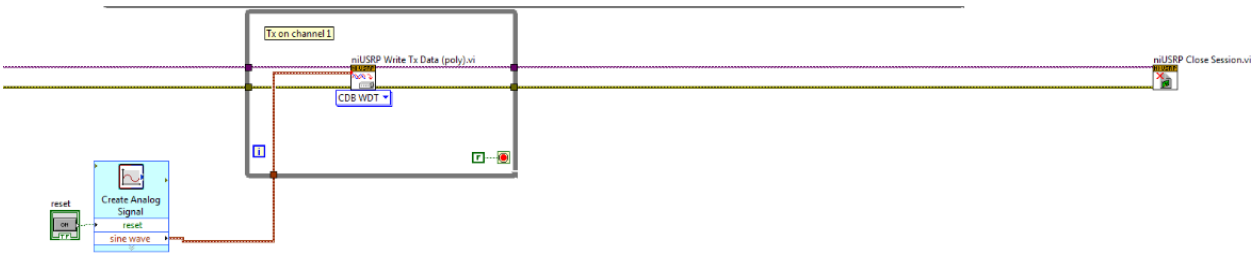


Figure 19: Tx session code

Similarly for receiving data we have to again initialize receiving session by using “niUSRP Open Rx Session .vi” and give session parameters like IQ rate (100000), Enabled Channel (Channel 0 for Rx), active antenna (Rx2) and gain. The code for this session is shown below:

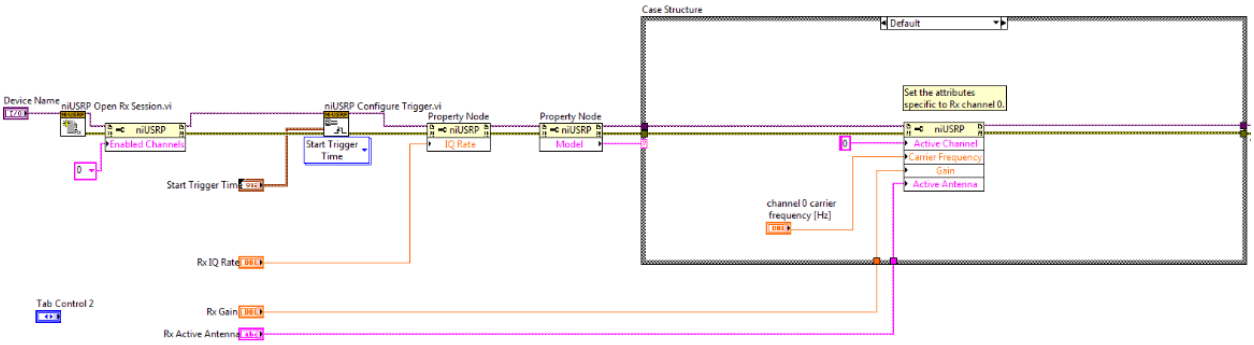


Figure 20: Initializing Rx session

After initializing receiving session “niUSRP Rx Fetch Data .vi” is used to fetch data or incoming signal from antenna and then we apply pour signal detection scheme to get our desired results. All of receiving session code is placed in a continuous while loop to continuously observe receiving data. We are displaying Fast Fourier transform graph and observe the peaks of the received signal using “Peak Detector .vi” block. We have added a

"Math Script" in our code to place check on the received signal's amplitude since, the amplitude of signal reflected by a metallic object is greater than other objects so we say that if the received amplitude is greater than our set threshold than the signal is reflected from a metallic object. The LabVIEW block diagram for this code is shown below:

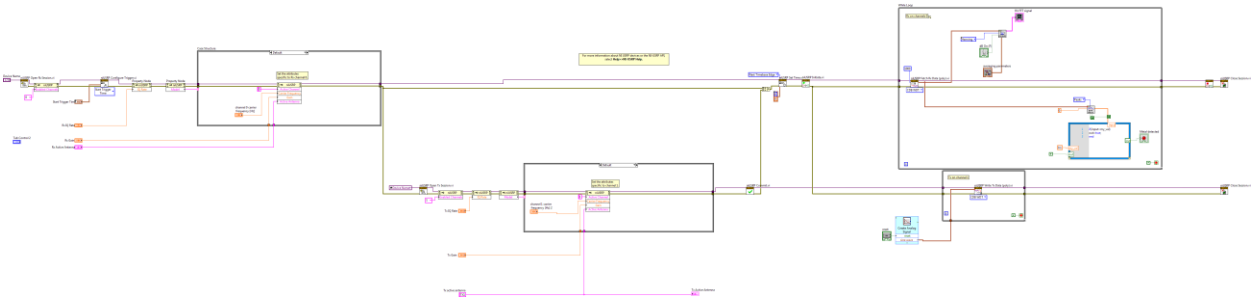
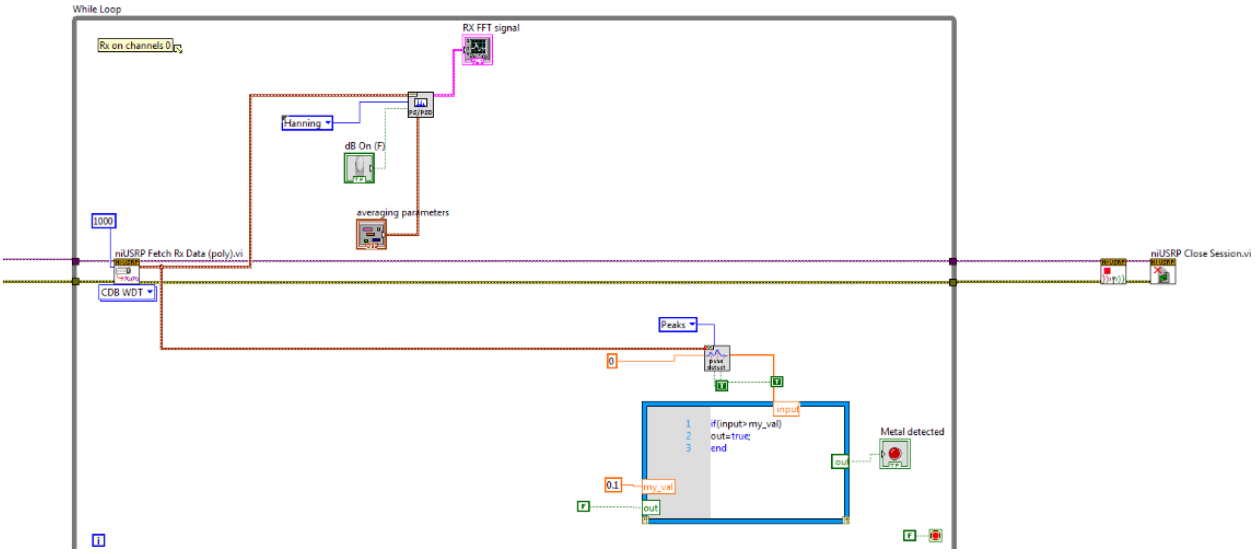


Figure 21: Complete Block Diagram



For complete application all above discussed blocks are compiled together in single block diagram as shown below:

Figure 22: Rx session code

CHAPTER 7. RESULTS AND ANALYSIS

7.1 Horn Antennas

The horn antennas we designed for the project were physically tested using Vector Network Analyser (VNA) and the following output (in figure below) was observed:

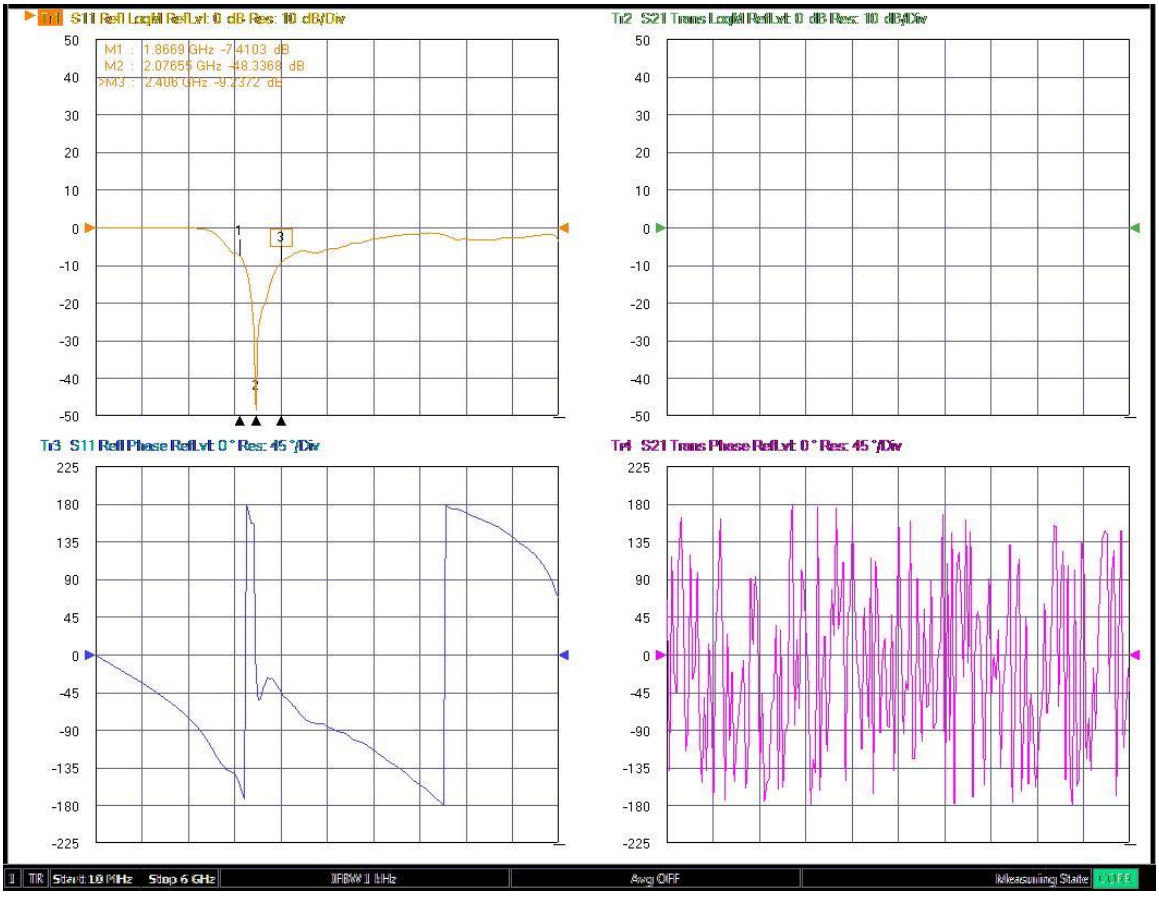


Figure 23: VNA outputs

The above figure shows the s-parameters of the horn antenna the most significant parameter for our analysis is the s-11 parameter as it shows the reflection or the range of the allowable frequency bandwidth of the antenna. The signal is said to be in allowable bandwidth of

antenna if the s-11 parameter at the signal's frequency is less than -10 dB therefore, we say all signals for which the s-11 parameter at their frequencies is less than -10 dB are allowed to pass. As shown by VNA results the allowable frequency range of our antenna is 1.9 GHz to 2.4 GHz with maximum reflection at 2.09 GHz around -50 dB. These results perfectly match with or software designed results which have been discussed above.

7.2 ***Metal detection in free space***

For metal detection in free space the system easily detects any object within the directive path of horn antennas and we can easily observe the change in reflected signals magnitude with high precision but the value of these magnitudes vary with the change of environment so we have to calibrate the threshold values placed on amplitude to define a metallic object whenever the system is moved from one environment to the other.

Once the device is calibrated the system accurately differentiates the presence of a metallic object (we used an aluminium sheet as reference) from other objects and we tested multiple objects like human body, wooden board, book, mobile phones etc.

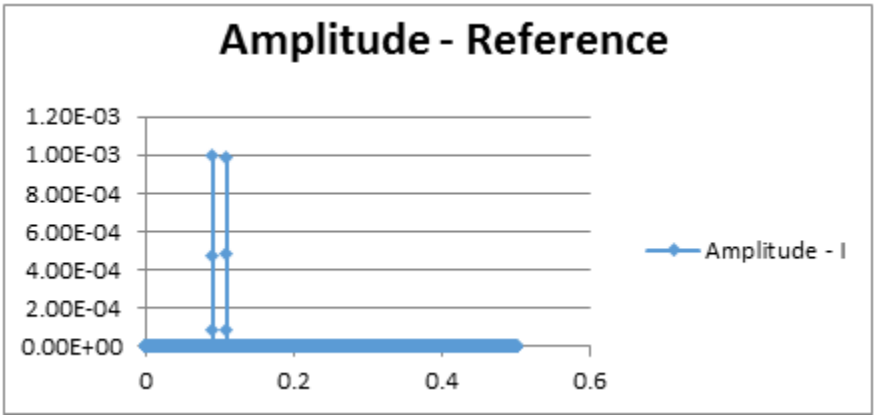
We also tested the system by creating a dummy form of suicide jacket by placing a metallic sheet in a human clothing to replicate the metallic contents of suicide vest which usually include metallic ball bearings and other metallic substances. We were successfully able to differentiate between the person wearing a vest and a person without a vest.

The problems encountered in this system is that if any object whether it is metallic or not passes swiftly in front of the antennas then there is a sudden change in the reflected signal's

amplitude due to Doppler Effect therefore, system identifies as if a metal is present in front of it.

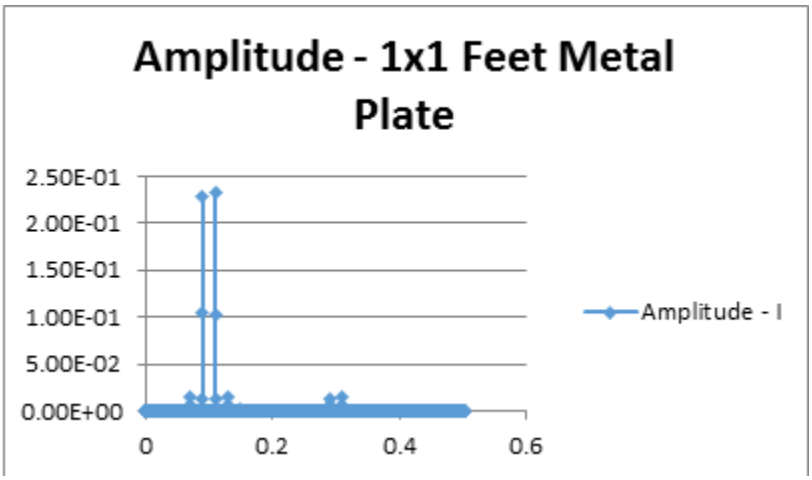
The other problem we encountered is that the system is unable to differentiate between a non-metallic object placed very close to it (around 10-20cm) and a metallic object placed far away (around 10 feet or more). This reason for this is the amplitude of reflected wave from a non-metallic object placed near to system is same as the amplitude of reflected signal from a metal placed far away due to signal attenuation and other factors that cause signal losses.

7.2.1 ***Testing with less than two feet distance***



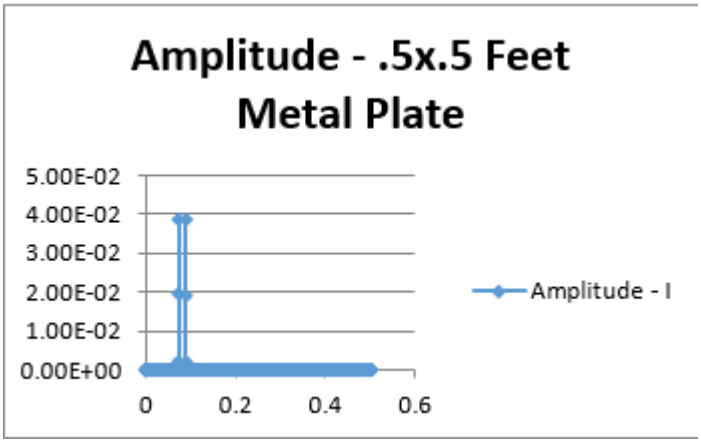
The above graph gives us the graph gives us the Fast Fourier Transform of the reflected wave and this can be considered as base or the reference value for the further testing.

7.2.1.1 *1x1 Feet Metal Plate*



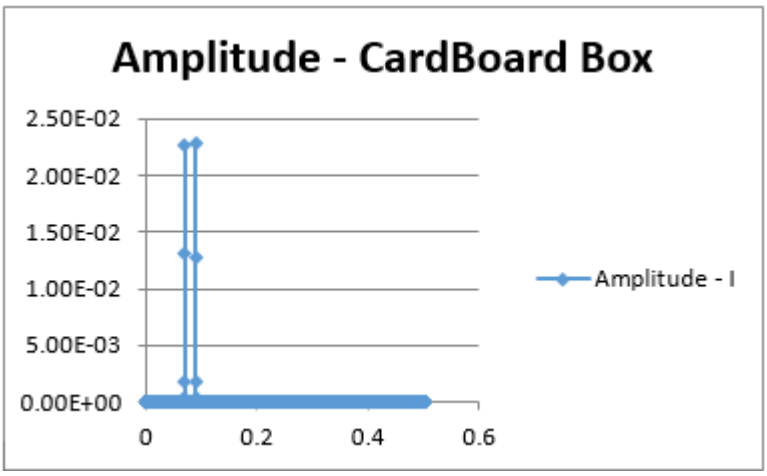
With the Introduction of the 1x1 feet metal plate the magnitude of the reflected wave is increased almost 200 times. This indicates the presence of the metal in the space.

7.2.1.2 *5x.5 Feet Metal Plate*



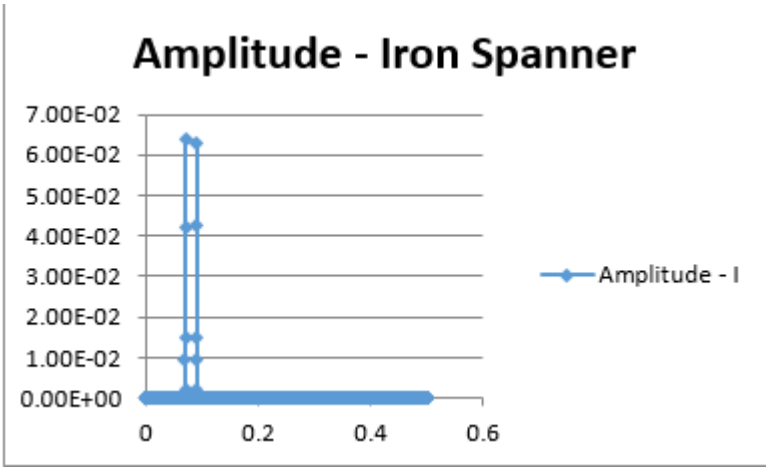
With the Introduction of the .5x.5 feet metal plate the magnitude of the reflected wave is increased almost 35 times as compared to the reference value. This indicates the presence of the metal in the space.

7.2.1.3 *A cardboard Box*



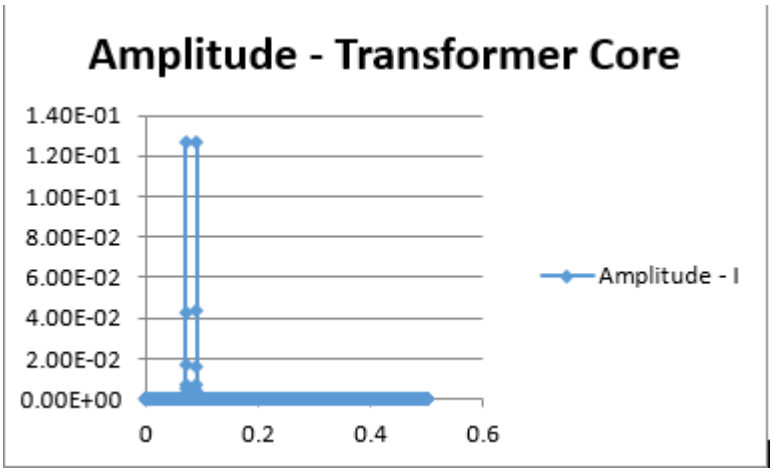
As we had to test other small metallic objects so we placed a cardboard box on which the metallic objects were placed. Now this peak value was taken as the reference for further testing of small metallic objects.

7.2.1.4 *An Iron Spanner Placed on the Cardboard Box*



With the Introduction of the iron spanner the magnitude of the reflected wave is increased from 0.02 to 0.06. This indicates the presence of the metal.

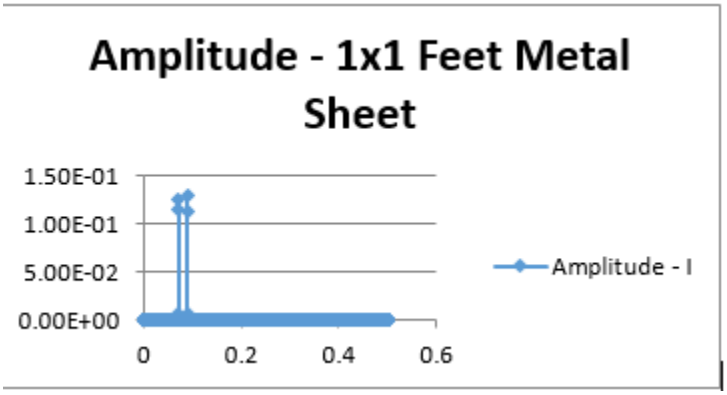
7.2.1.5 *Transformer Core Placed on the Cardboard Box*



With the Introduction of the Transformer core the magnitude of the reflected wave is increased from 0.02 to 0.12. This indicates the presence of the metal.

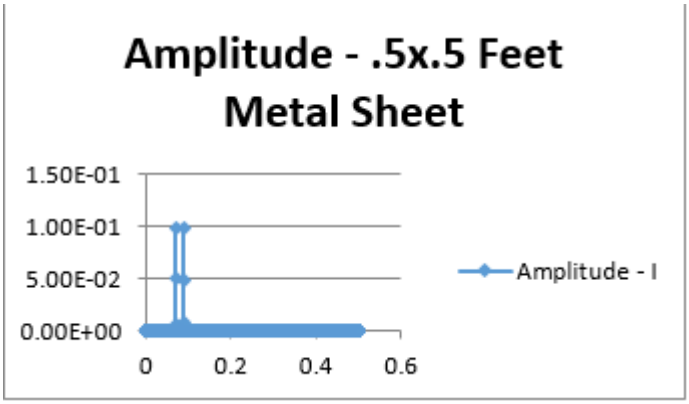
7.2.2 ***Testing with four feet distance***

7.2.2.1 *1x1 Feet Metal Sheet*



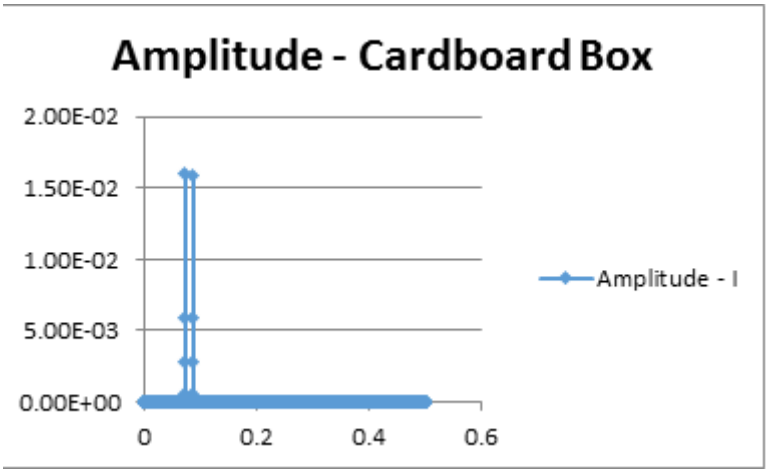
With the Introduction of the 1x1 feet metal plate the magnitude of the reflected wave is increased almost 120 times. This indicates the presence of the metal in the space.

7.2.2.2 *5x5 Feet Metal Sheet*



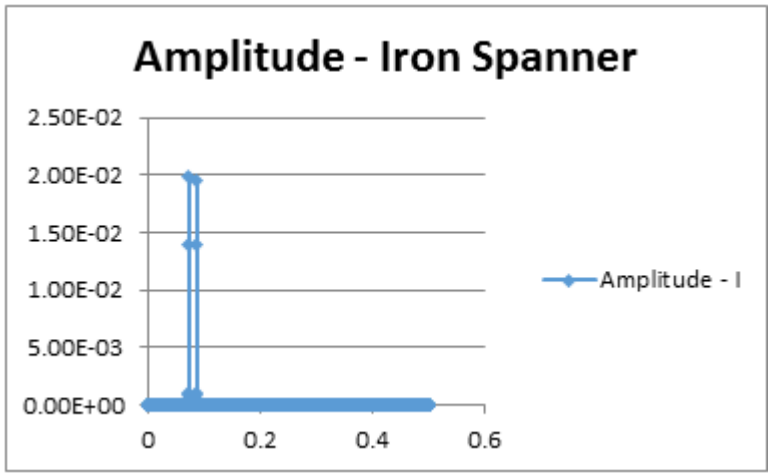
With the Introduction of the .5x.5 feet metal plate the magnitude of the reflected wave is increased when compared to the reference value. This indicates the presence of the metal in the space.

7.2.2.3 Cardboard Box



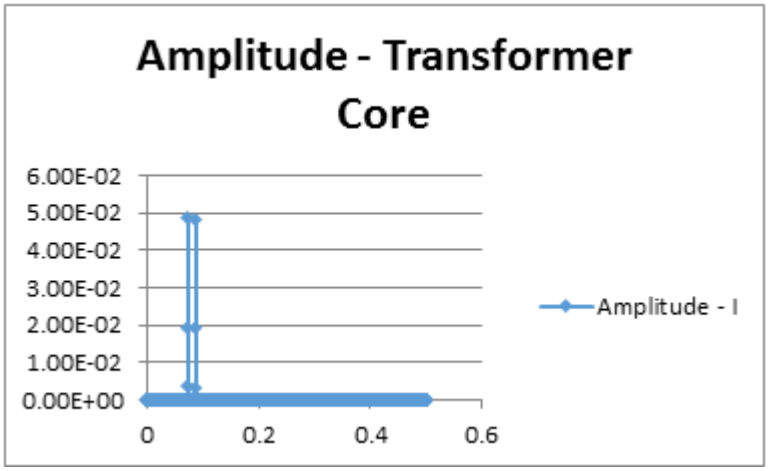
As we had to test other small metallic objects so we placed a cardboard box on which the metallic objects were placed. Now this peak value was taken as the reference for further testing of small metallic objects.

7.2.2.4 An Iron Spanner



With the Introduction of the iron spanner the magnitude of the reflected wave is increased from 0.016 to 0.02. This indicates the presence of the metal.

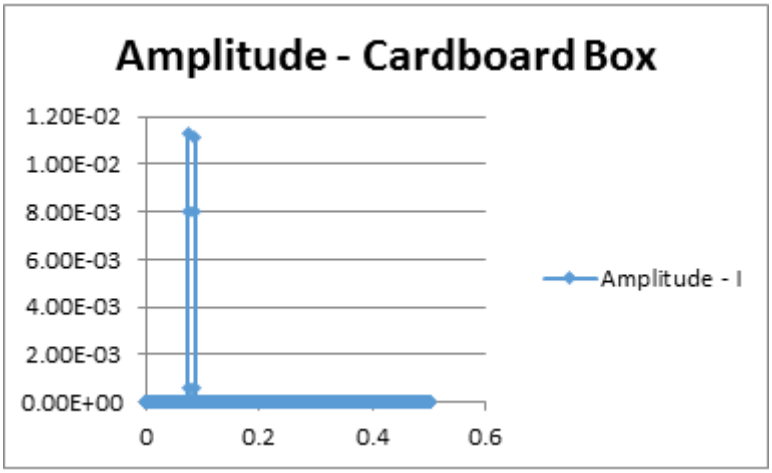
7.2.2.5 Transformer Core Placed on the Cardboard Box



With the Introduction of the transformer core the magnitude of the reflected wave is increased from 0.016 to 0.05. This indicates the presence of the metal.

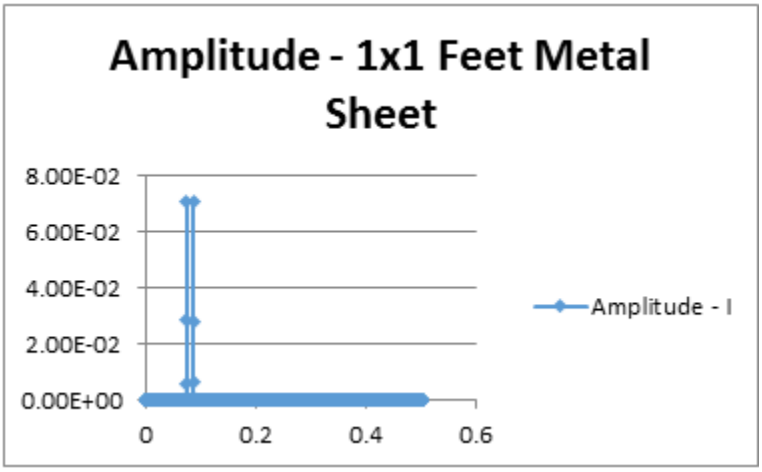
*7.2.3 **Testing with ten feet distance***

7.2.3.1 Cardboard Box



As we had to test the presence of metal at long range so we placed a cardboard box on which the metallic was attached. Now this peak value was taken as the reference for further testing.

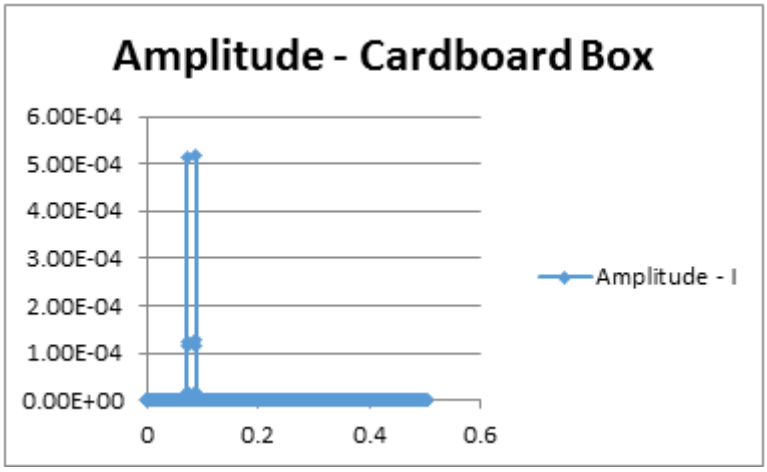
7.2.3.2 1x1 Feet metal Sheet



With the Introduction of the 1x1 metal sheet the magnitude of the reflected wave is increased from almost 0.01 to 0.07. This indicates the presence of the metal.

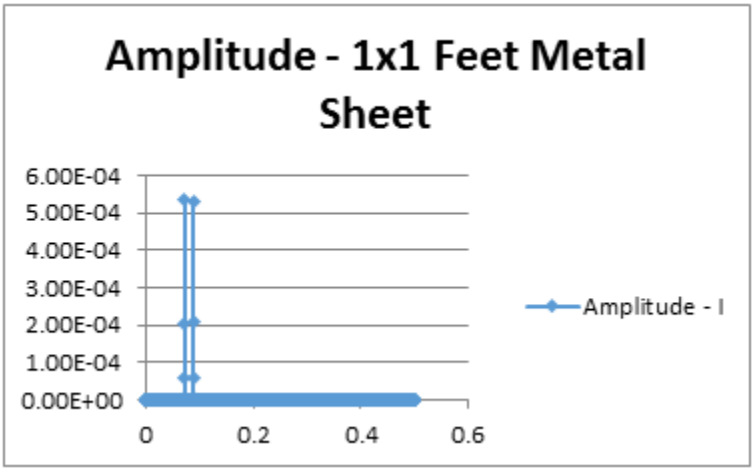
*7.2.4 **Testing with twenty feet distance***

7.2.4.1 Cardboard box



As we had to test the presence of metal at long range so we placed a cardboard box on which the metallic was attached. Now this peak value was taken as the reference for further testing.

7.2.4.2 1x1 Feet metal Sheet



Even though there was not any significant change in the reflected waves magnitude at 20 feet distance but a slight increase was somehow observed and this indicates that there is a metal present.

*7.3 **Ground Penetrating Radar (GPR)***

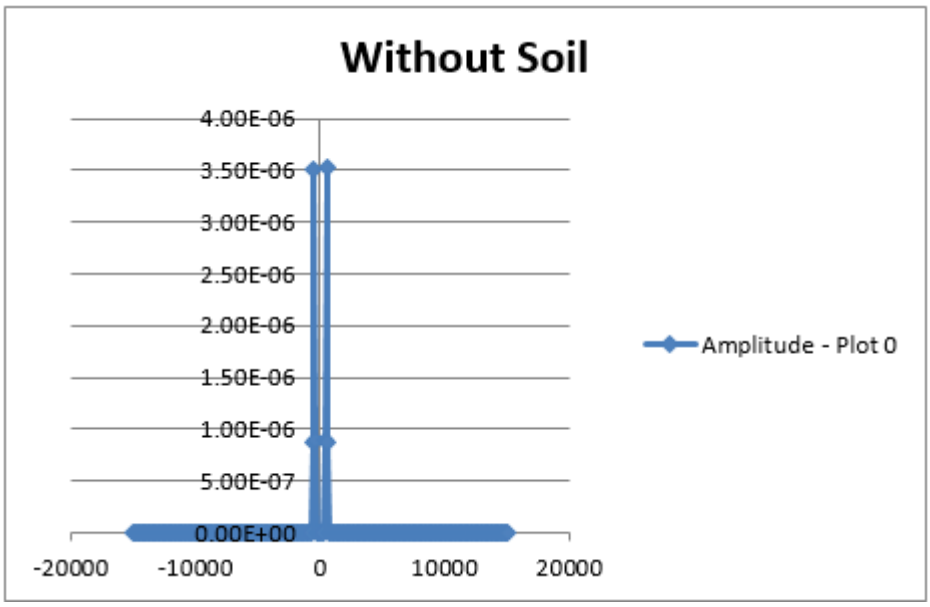
For GPR the full-duplex scheme works fine and the readings vary when the system is moved over clay and when a metallic sheet is placed under the clay. The accurate depth for the GPR is found to be around 2 feet under the artificial environment created in the Lab.

The thresholds set on received signal's amplitude to differentiate whether a metal is placed under ground or not because the values vary because USRP is not a calibrated device. Therefore we were unable to set different thresholds to differentiate between different minerals present under the ground. Also, the full-duplex scheme used has some computing delay due to software's computing limitations therefore, for accurate measurements GPR should be moved very slowly.

*7.3.1 **GPR Testing in dry soil***

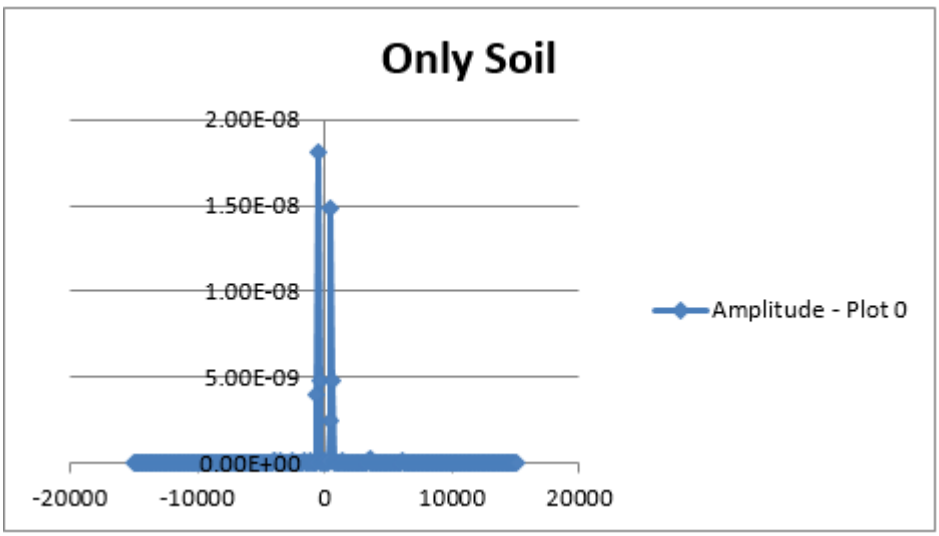
7.3.1.1 Test in free space:-

First of all we tested the GPR system pointed downward to ground in air and noted the reflected signal's FFT graph which is shown below:



7.3.1.2 Reference Test in dry soil:-

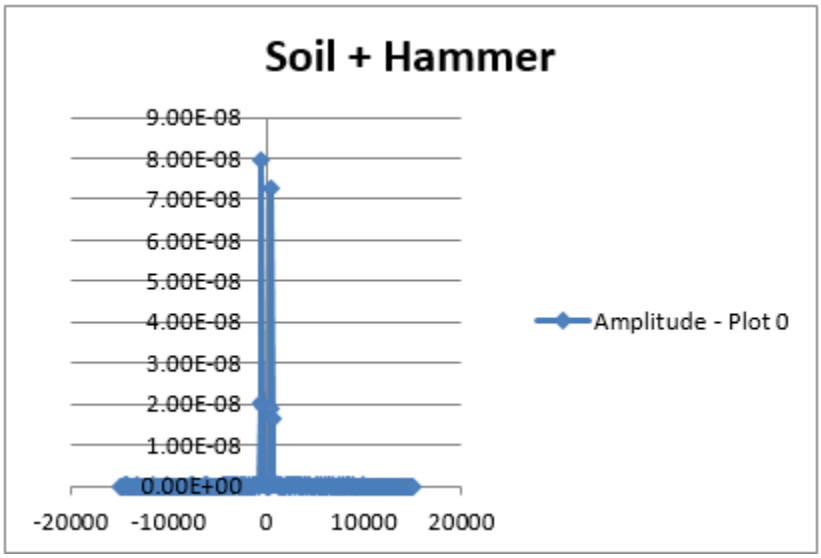
Then we placed a plastic container on the floor filled with mud/soil and reflected signal is shown in the figure below. We took these readings as our base readings for all other tested objects placed underneath the soil.



In only soil the FFT value of reflected soil comes out to be 1.5e-8 to 1.75e-8. We had taken these values as base value for further testing with different materials.

7.3.1.2 Hammer under dry soil:-

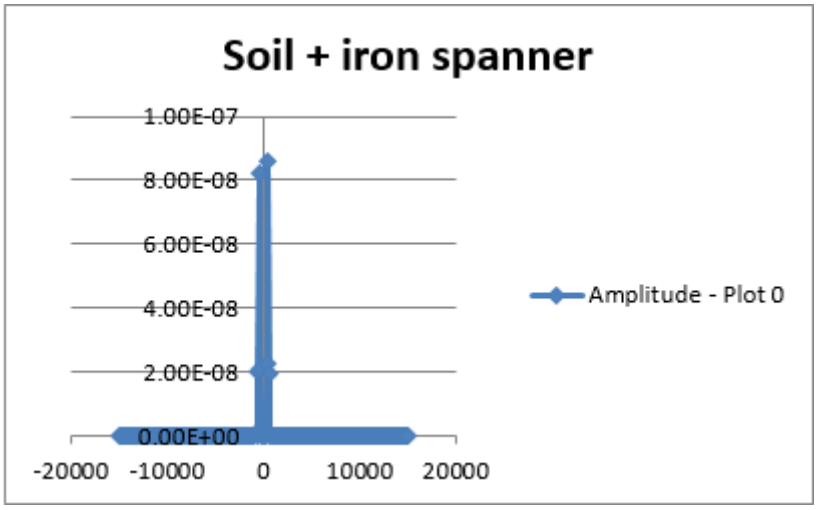
We repeated above experiment by placing hammer under the soil and the resulting graphs is shown below:



As we can observe in the above graph the magnitude of the reflected wave increased from 1.75e-8 to 8.0e-8 as a metal is under the soil. These values would help us in setting values for predicting different types of metals under soil.

7.3.1.3 Iron Spanner under dry soil:-

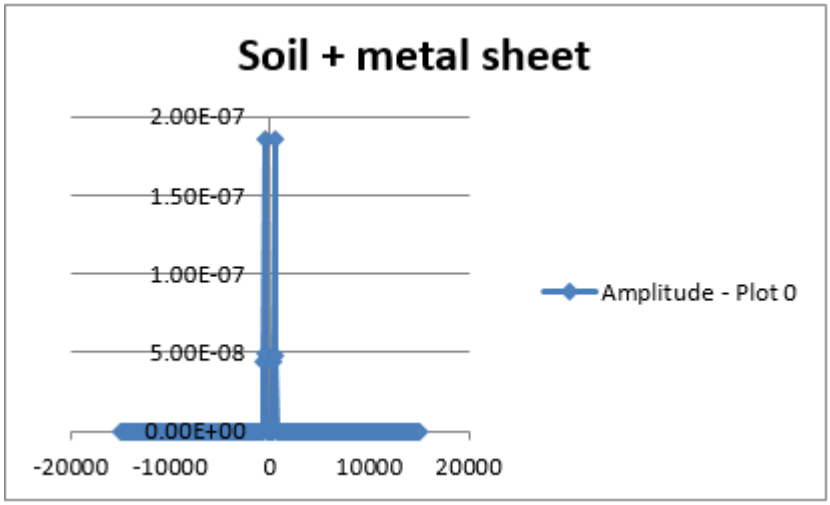
We repeated the above experiment by placing iron spanner under soil and graph is shown below.



The magnitude of the reflected wave is a little bit more than in the previous experiment. The difference is quite enough for differentiating this object from previous.

7.3.1.4 Metal Sheet under dry soil:-

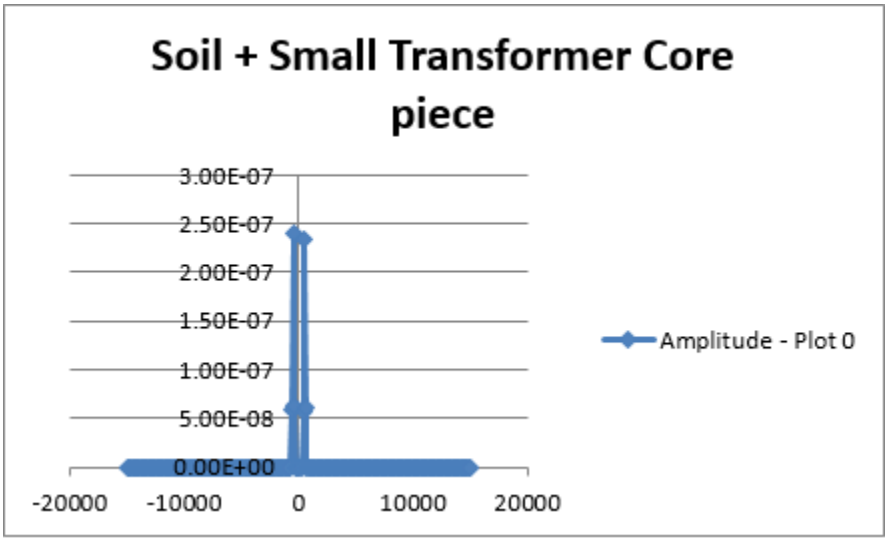
We repeated the earlier experiment by placing metal sheet under soil. The result shown below.



This time the increase in the magnitude of the reflected wave is quite a large which is 1.75e-7. This sharp increase is due the highly reflected surface.

7.3.1.5 Small transformer core under dry soil:-

The last object used for testing is a small piece of transformer core. The result of the FFT plot are shown below

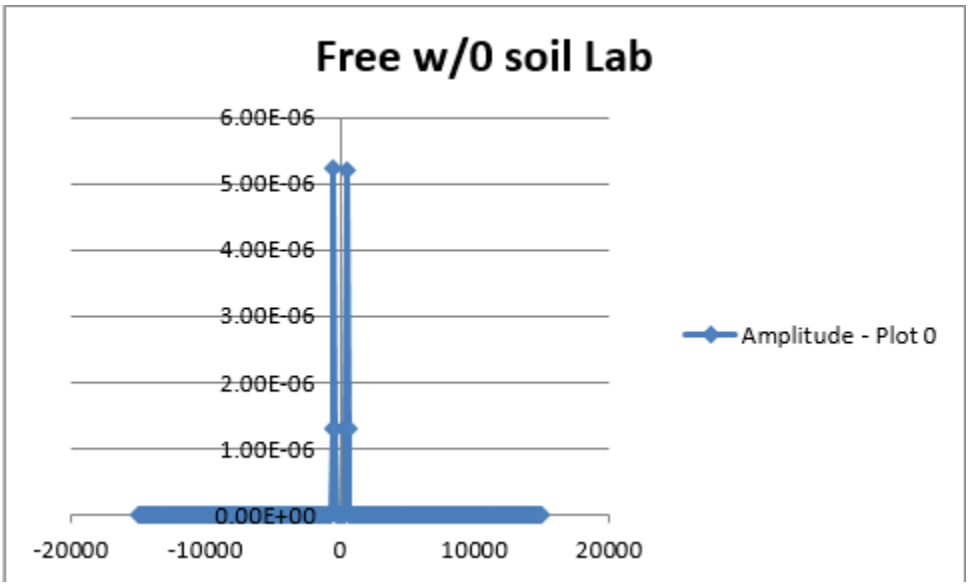


The magnitude of the reflected wave is the highest among all the values. But its difference from the metal sheet is not very high the metal sheet had the highest magnitude of 1.8e-7 and its magnitude is 2.4e-7. But the difference could be easily differentiated.

*7.3.2 **GPR Testing in wet soil***

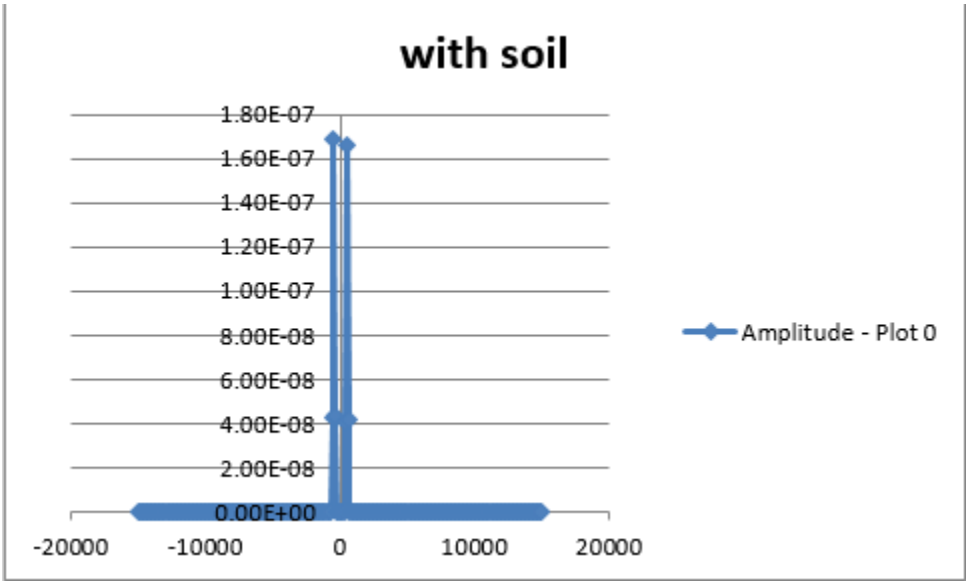
7.3.2.1 Test in free space:-

First of all we tested the GPR system in our lab pointed downward to ground in air and noted the reflected signal's FFT graph which is shown below:



7.3.2.2 Reference Test in wet soil:-

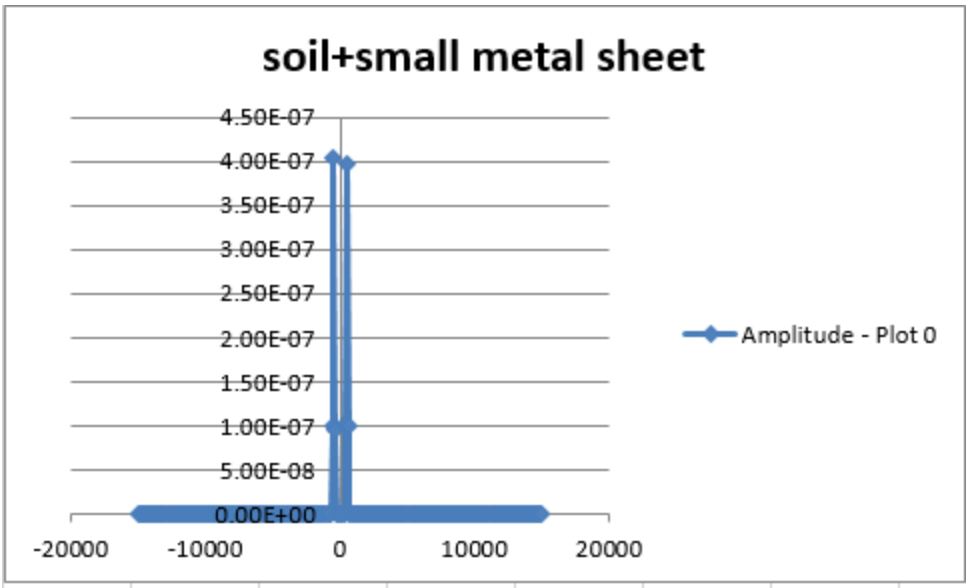
Then we placed a plastic container on the floor filled with wet mud/soil and reflected signal is shown in the figure below. We took these readings as our base readings for all other tested objects placed underneath the wet soil.



In only soil the FFT value of reflected soil comes out to be 1.6e-7. This value is large then for the value of dry soil that was 1.5e-8. This means that the wet soil increase the signal strength. We had taken these values as base value for further testing with different objects.

7.3.2.2 Small metal sheet under wet soil:-

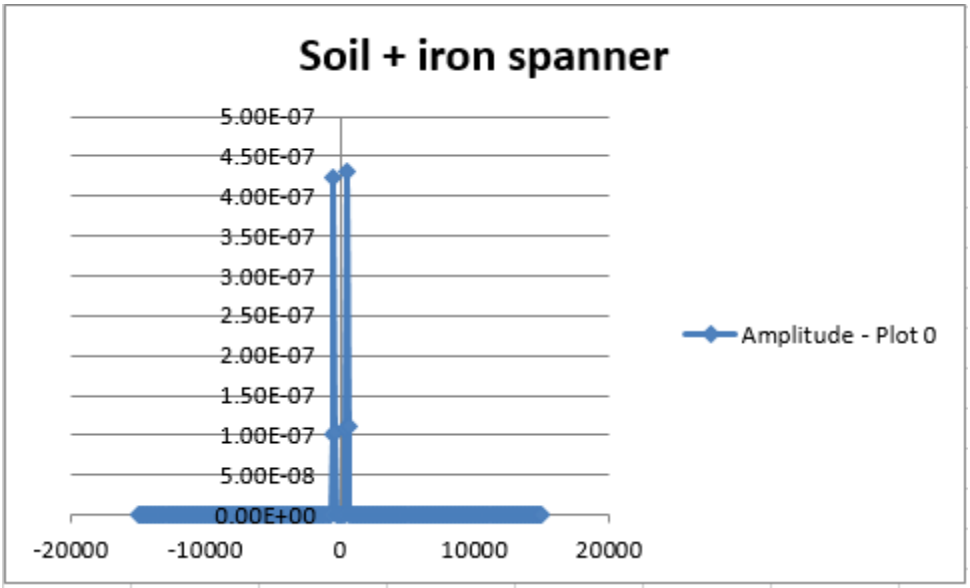
We repeated above experiment by placing small metal sheet under the soil and the resulting graphs is shown below:



As we can observe in the above graph the magnitude of the reflected wave increased from 1.6e-7 to 4.0e-8 as a metal is under the soil. These values would help us in setting values for predicting different types of metals under wet soil.

7.3.2.3 Iron Spanner under wet soil:-

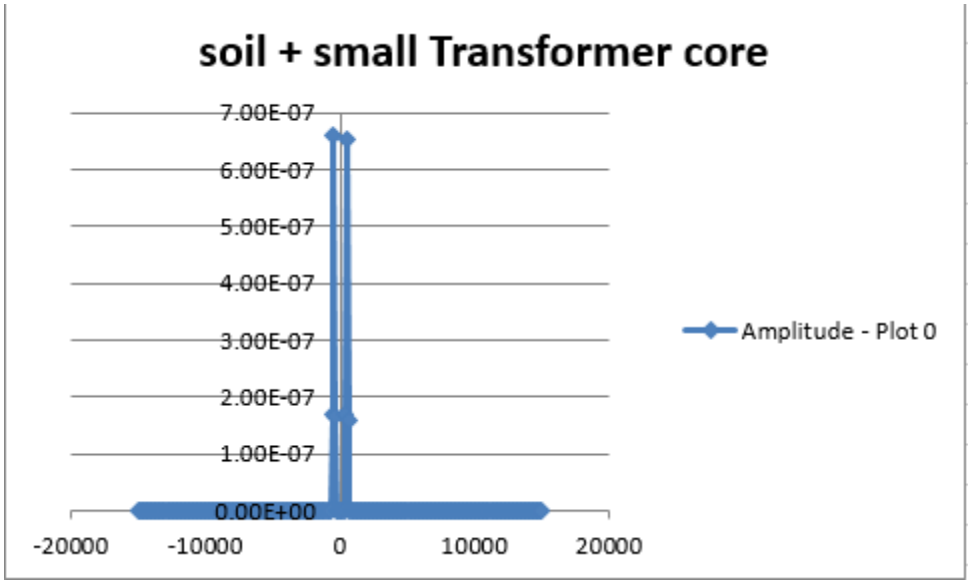
We repeated the above experiment by placing iron spanner under wet soil and graph is shown below.



The magnitude of the reflected wave is a little bit more than in the previous experiment. It was increased from 4.0e-7 to 4.3e-7. The difference is quite enough for differentiating this object from previous.

7.3.2.4 Small transformer core under wet soil:-

The last object used for testing is a small piece of transformer core. The result of the FFT plot are shown below



The magnitude of the reflected wave is the highest among all the values. But its difference from the iron spanner is not very high. The iron spanner had the highest magnitude of 4.3e-7 and its magnitude is 6.7e-7. But the difference could be easily differentiated.

CHAPTER 8. CONCLUSION

We were able to achieve most of our proposed objectives with some limitations which mainly aroused due to hardware limitations. The most significant achievement from this project is the software designing of horn antennas and their accurate hardware implementation. The designed antennas provide highly accurate results and can be used for other communication and radio frequency applications. The second objective was to find an accurate method to differentiate between a metal and non-metal in free space to detect a weapons and suicide vests to ensure safety of public places and other security central places. The same scheme was implemented in form of GPR used to detect underground pipes, electrical wiring and even pipe leakages etc. The project was divided into both research and hardware development to find a definite method to ensure public safety but after considerate research and intuition we implemented our research findings to develop a GPR which can be further improved. Finally, we were able to implement our findings in practically and designed not only one but two hardware prototypes and two high precision horn antennas.

CHAPTER 9. RECOMMENDATIONS

Our own designed and physically made horn antenna are quite good. They give a very good db. Gain value. We have done a proof of concept of a very important and demanding product that is long range metal detector. In future we will try to make it a product which actually involves a little bit of hard work with calibrated hardware. A highly calibrated hardware that can give us a very accurate magnitude as well as phase of the reflected waves that will eventually help in making a product which could save lives of many people. Similarly our GPR would also get improve if we use a calibrated SDR for our GPR with antennas that can help large frequency range and high directivity.

CHAPTER 10. REFERENCES

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